

Efficient and Secure Embedding Transmission for Edge–Cloud Split Vision–Language Models: A System-Level Security Evaluation

Introduction

Vision–Language Models (VLMs) are increasingly used in multimodal AI applications. However, their computational demands make it difficult to deploy them fully on resource-limited edge devices, often requiring raw data to be transmitted to the cloud for processing. A distributed VLM architecture mitigates this limitation by computing intermediate embeddings on edge devices and transmitting them to the cloud. Although this architecture avoids direct transmission of raw input data, embedding transmission still poses privacy challenges that require appropriate protection under edge resource limits. Therefore, this project evaluates secure transmission mechanisms for visual embeddings in distributed VLM systems, focusing on security-efficiency trade-offs in realistic deployment scenarios.

Background Research or Context

Recent advances in multimodal pretraining have driven the development of Vision–Language Models (VLMs), which integrate visual and textual inputs within a unified framework [1]. Li et al. [2] proposed a distributed VLM architecture in which early layers are computed on resource-limited edge devices, and intermediate visual embeddings are transmitted to the cloud for further processing. This approach reduces local computational demands and network traffic, while avoiding directly transmitting raw visual data. However, despite these advantages, the security of intermediate representations remains a critical concern. Pham et al. [3] show that intermediate visual embeddings are not inherently privacy-preserving, as reconstruction and inference attacks can recover sensitive information under certain threat models. However, existing studies primarily focus on leakage analysis, and pay limited attention to the system-level evaluation of secure transmission mechanisms under realistic edge resource limits.

Proposed Project

This project focuses on evaluating secure transmission mechanisms for intermediate visual embeddings in distributed VLM systems. The workflow consists of three main components: (1) implementing a baseline distributed VLM system to emulate an edge–cloud deployment; (2) integrating and comparing secure transmission mechanisms, including transport encryption, integrity protection, and authentication schemes, to evaluate their computational overhead and security strength; and (3) conducting quantitative system-level trade-off analysis by comparing bandwidth consumption, latency overhead, and computational cost before and after the integration of secure transmission mechanisms under realistic edge constraints.

Timeline

The project consists of four stages. Stage 1 (Mar 1–Jun 6) focuses on literature review, experimental design, dataset and model selection, and experimental environment setup. Stage 2 (Jun 7–Jul 3) implements a baseline distributed VLM system to emulate an edge–cloud deployment. Stage 3 (Jul 4–Jul 20) implements and compares multiple secure transmission strategies for intermediate embeddings, measuring their impact on CPU utilization, network throughput, and latency relative to the baseline system. Stage 4 (Jul 21–Aug 20) analyzes experimental results to quantify system-level security–efficiency trade-offs, alongside dissertation writing and final revision.

References

- [1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh *et al.*, “Learning transferable visual models from natural language supervision,” in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.
- [2] Y. Li, D. Gumaste, M. K. Turkcan, J. Ghaderi, G. Zussman, and Z. Kostic, “Distributed vlms: Efficient vision-language processing through cloud-edge collaboration,” in *2025 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*. IEEE Computer Society, 2025, pp. 280–286.
- [3] N. D. Pham and N. Chilamkurti, “Data leakage threats and protection in split learning: a survey,” in *Proceedings of the 2023 International Conference on Intelligent Computing and Its Emerging Applications*, 2023, pp. 141–147.